



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-18

Experiment 18. Student-Teacher-Subject

Aim

To write a Prolog program to represent the relationship between students, teachers and subjects.

Objectives

- To create relationships using Prolog facts.
- To understand database querying.
- To retrieve teacher information for a student.

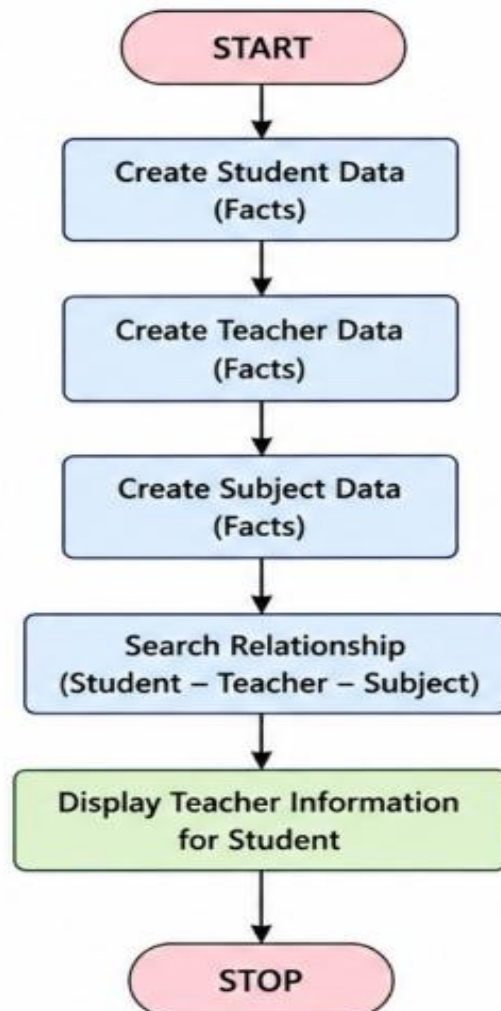
Algorithm

1. Define student and subject facts.
2. Define teacher and subject facts.
3. Create a relationship between students and teachers.
4. Query the required student.
5. Display the teacher.

Flowchart

18. STUDENT-TEACHER-SUBJECT

FLOWCHART



Code

student(rahul, ai).

student(priya, ml).

student(anu, python).

```
teacher(kumar, ai).
```

```
teacher(rajesh, ml).
```

```
teacher(suresh, python).
```

```
student_teacher(Student,  
Teacher) :-
```

```
    student(Student,  
Subject),
```

```
    teacher(Teacher,  
Subject).
```

```
main :-
```

```
    Student = rahul,
```

```
    student_teacher(Student,  
Teacher),
```

```
    write('Student: '),  
write(Student), nl,
```

```
    write('Teacher: '),  
write(Teacher), nl.
```

```
:- initialization(main).
```

Output

```
HelloWorld.pl x
```

```
student(rahul, ai).
student(priya, ml).
student(anu, python).

teacher(kumar, ai).
teacher(rajesh, ml).
teacher(suresh, python).

student_teacher(Student, Teacher) :-
    student(Student, Subject),
    teacher(Teacher, Subject).

main :-
    Student = rahul,
    student_teacher(Student, Teacher),
    write('Student: '), write(Student), nl,
    write('Teacher: '), write(Teacher), nl.

:- initialization(main).
```

I/O AI Agent

STDIN

Output:
Student: rahul
Teacher: kumar

Result

Student= Rahul

Teacher = kumar.

Conclusion

Thus, the student-teacher-subject relationship was successfully implemented using Prolog.